Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

Topic 6: Security, Privacy and Data Integrity

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 5 minutes

TOTAL MARKS: 56

Marks obtained: ..... / 56

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 56.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 11 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.
· Only 56 marks of past-paper questions are available for this chapter, so the recommended time is 1 hour 5 minutes (1

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 A shop rents cars to customers. The shop uses a relational database to store information about the rentals.

(a) The data is validated and verified when it is entered into the database.

(i) The car registration number must be: 1 letter, followed by 3 numbers, followed by 2 letters.

For example, A123AA is valid but A12AA is invalid.

One way that a registration number can be validated is by using a presence check to make sure the registration number has been entered.

Describe **two other** ways that the car registration number can be validated.

1 .....

.....

2 .....

.....

[2]

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(ii) Describe **two** ways that the car registration number can be verified when it is entered into the database.

1 .....

.....

2 .....

.....

[2]

(iii) State why the car registration number might be incorrect even after it has been validated and verified.

.....

..... [1]

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2 An assessment board scans exam papers and stores the digitised papers on a server. Exam markers download the digitised papers to mark. The exam markers then upload the mark for each paper.

(a) The assessment board needs to make sure the data stored on the server is secure.

(i) Authentication methods can help to protect the server against hackers.

Identify **one other** security measure that helps to protect the server from hackers.

Describe how the security measure works.

Security measure .....

Description .....

.....

.....

.....

[3]

(ii) Identify **one** security measure that helps to protect the data when it is being transmitted to its destination. Describe how the security measure works.

Security measure .....

Description .....

.....

.....

.....

[3]

3 Personal data is entered into a computer using a keyboard. The data is validated and verified during data entry.

(a) Describe **two** methods of data verification that can be used during data entry.

1 .....

.....

.....

2 .....

.....

.....

[4]

(b) The personal data includes the date of birth, for example 05/07/2010.

Complete the table by identifying **two** methods of validation that can be used on the date of birth.

State how each validation method will be used for the date of birth.

data validation method	use
....	....
....	....

[3]

(c) The data needs to be kept secure whilst on the computer.

(i) A virus is a threat to data.

Complete the table by describing what is meant by a virus.

Identify **and** describe **one** method to restrict the risk posed by a virus.

virus description	....
method to restrict risk	.....
description of method to restrict risk	....

[3]

(ii) A hacker is another threat to data.

Complete the table by describing what is meant by a hacker.

Identify **and** describe **one** method to restrict the risk posed by a hacker.

hacker description	....
method to restrict risk	.....
description of method to restrict risk	....

[3]

4 Validation and verification can be used to protect the integrity of data.

Complete the description of methods of ensuring data integrity by writing the missing terms.

..... is a method of verification where the data is entered by a user and then manually compared to the original data to make sure that they are the same.

A verification method used during data transfer calculates a value from the data which is transferred with the data. The receiver performs the same calculation and compares the result with the value transferred.
This method of verification is ..... .

There are several methods of data validation. ..... check compares the data entered to a list of possible values that can be entered. If the data is not one of these values it is rejected.

..... check can be used to make sure data entered has the correct combination of letters, numbers and symbols. For example, a code may need to be two capital letters, one number followed by one symbol.

..... check is used to make sure the correct quantity of characters has been entered. For example, a name must be more than one character and less than or equal to 100 characters.

[5]

5 **(a)** State the meaning of **privacy of data**.

.....

..... [1]

(b) State the meaning of **integrity of data**.

.....

..... [1]

(c) Describe the following threats to a computer system.

Phishing email .....

.....

.....

.....

Spyware .....

.....

.....

.....

[4]

6 A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

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(a) An example of a tutor ID is NK16C6.

An administrative officer enters the tutor ID into the `TUTOR` table.

Explain how data verification can be used when the tutor ID is entered.

.....

.....

.....

.....

.....

.....

.....

..... [4]

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7 **(a)** Data verification is one method of protecting the integrity of data.

Describe **one** other method of protecting the integrity of data.

.....

.....

.....

..... [2]

(b) State **one** difference and **one** similarity between pharming and phishing.

Difference .....

.....

Similarity .....

.....

[2]

(c) Explain how the data security risks of malware can be restricted.

.....

.....

.....

.....

.....

..... [3]

8 A program is written in a high-level language by a team of three programmers using an Integrated Development Environment (IDE).

(a) The file containing the final program code will be sent by email for beta testing.

Identify **one** security method that can be used to protect the program code from unauthorised access during email transfer.

Explain how your chosen method protects the program code.

Security method .....

.....

Explanation .....

.....

.....

.....

[3]

9 Draw **one** line from each verification method to indicate whether it is used during data transfer or data entry.

Verification Method	Use
Parity byte check	
Checksum	Data transfer
Visual check	
Parity block check	Data entry

[2]

10 (a) A computer system uses even parity. The least significant (rightmost) bit of each byte is the parity bit.

(i) Complete the byte by writing the missing parity bit:

							parity bit ↓
0	1	0	1	1	1	0	

[1]

(ii) The computer also uses parity block check. The parity block check uses even parity. Computer A transmits four bytes of data to computer B, followed by a parity byte. Computer B receives the following sequence of bytes.

								parity bit ↓
	1	0	1	1	0	1	1	1
	0	1	1	1	0	0	0	0
	0	0	0	1	1	0	1	1
	0	1	1	1	0	1	0	0
parity byte →	1	0	1	0	0	0	0	0

Following transmission, one of the four bytes of data has an error in one of the bits.

Circle the bit that has been altered during the data transfer. [1]

11 A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

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(a) The table contains three algorithms that perform data validation. The third algorithm uses the functions `MID` and `LENGTH`.

The functions `MID` and `LENGTH` are defined as follows:

- `MID(ThisString : STRING, x : INTEGER, y : INTEGER) RETURNS STRING` returns a string of length `y` starting at position `x` from `ThisString`
- `LENGTH(ThisString : STRING) RETURNS INTEGER` returns the integer value representing the length of `ThisString`

Complete the table to identify the method of data validation for each algorithm. Each method of data validation must be different.

Algorithm	Method of data validation

`INPUT x`
 	
`IF x < 1 or x > 26 THEN`
 	
`  OUTPUT "Invalid"`
 	
`ENDIF`	

`INPUT x`
 	
`IF x <> 'R' AND x <> 'G' AND x <> 'B' THEN`
 	
`  OUTPUT "Invalid"`
 	
`ENDIF`	

`INPUT x`
 	
`FLAG ← FALSE`
 	
`FOR INDEX ← 1 TO LENGTH(x)`
 	
`  IF MID(x, INDEX, 1) = "@" THEN`
 	
`  FLAG ← TRUE`
 	
`  ENDIF`
 	
`NEXT INDEX`
 	
`IF NOT FLAG THEN`
 	
`  OUTPUT "Invalid"`
 	
`ENDIF`	

[3]

END OF QUESTION PAPER

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